# Week 3 APP DEV-2

## 2024 Dec01 Quiz-2

**1.** [Source Q17; MSQ]

Consider the below Vue component.

```
<template>
  <div>
    <button @click="setDarkMode">Dark Mode</button>
    <button @click="setLightMode">Light Mode</button>
  </div>
</template>

<script>
export default {
  data() {
    return {
      theme: localStorage.getItem('theme') || 'light'
    };
  },
  methods: {
    setDarkMode() {
      this.theme = 'dark';
      localStorage.setItem('theme', 'dark');
    },
    setLightMode() {
      this.theme = 'light';
      localStorage.setItem('theme', 'light');
    }
  }
};
</script>
```

Which of the following statement(s) is/are correct about the behavior of the above component?

(a) The theme will persist across page reloads because it is stored in localStorage.

(b) If the user opens a new tab, the theme will reset to 'light' in that new tab.

(c) The theme will reset to 'light' every time the page is refreshed.

(d) The theme will be shared across all open tabs, as localStorage is accessibleacross tabs of the same origin.

[Source ans: a, d]

## 2024 Dec22 ET

**2.** [Source Q8; MCQ]

Consider the following JavaScript code.

```
function addItemToCart(item) {
        let cart = JSON.parse(localStorage.getItem('cart')) || [];
        cart.push(item);
        localStorage.setItem('cart', JSON.stringify(cart));
    }

    function getCartItems() {
        return JSON.parse(localStorage.getItem('cart')) || [];
    }

    addItemToCart({ id: 1, name: 'Laptop' }.name);
    console.log(getCartItems());
```

The above code is initially loaded on the browser and then the browser is refreshed two times. What will be the final output?

(a) []

(b) ['Laptop', 'Laptop', 'Laptop']

(c) [{ id: 1, name: 'Laptop' }]

(d) [undefined]

[Source ans: b]

**3.** [Source Q22; MCQ]

Which of the following statements is FALSE?

(a) Cache-Control: no-store prevents caching

(b) ETag helps validate cache freshness

(c) Data stored in sessionStorage remains available even after the browser isclosed and re-opened.

(d) localStorage has larger storage limit than cookies

[Source ans: c]

**4.** [Source Q24; MCQ]

Consider the below HTML document.

```
<!DOCTYPE html>
<html>
<body>
    <div>
        <div id="grandparent">GrandParent
            <div id="parent">Parent
                <div id="child"> Child</div>
            </div>
        </div>
    </div>
    <script>
        const grandParent = document.getElementById("grandparent");
        const parent = document.getElementById("parent");
        const child = document.getElementById("child");

        grandParent.addEventListener("click", (e) => {
            console.log("GrandParent");
        }, { capture: true });

        parent.addEventListener("click", (e) => {
            console.log("Parent");
        }, { capture: true });

        child.addEventListener("click", (e) => {
            console.log("Child");
        }, { capture: true });
    </script>
</body>
</html>
```

What will be logged on the console, if the user opens the rendered web page in a browser and clicks the text "Child"?

(a) ChildParent GrandParent

(b) Child

(c) GrandParentParent Child

(d) None of these

[Source ans: c]

## 2025 Mar16 Quiz-2

**5.** [Source Q4; MCQ]

What distinguishes a Progressive Web App (PWA) from a traditional web application?

(a) PWAs must be downloaded from an app store

(b) PWAs only work on mobile devices

(c) PWAs can work offline and be installed on the home screen

(d) PWAs require native mobile code

[Source ans: c]

**6.** [Source Q16; MCQ]

What will be the visited pages history when the user navigates between pages and refreshes the page?

Consider the below Vue.js application.

```
<template>
  <div>
    <button @click="goToPage('Home')">Go to Home</button>
    <button @click="goToPage('About')">Go to About</button>
    <button @click="goToPage('Contact')">Go to Contact</button>

    <div v-if="visitedPages.length">
      <h3>Visited Pages:</h3>
      <ul>
        <li v-for="(page, index) in visitedPages" :key="index">{{ page
}}</li>
      </ul>
    </div>
  </div>
</template>

<script>
export default {
  data() {
    return {
      visitedPages: []
    };
  },
  mounted() {
    const storedPages = JSON.parse(sessionStorage.getItem('visitedPages'))
|| [];
    this.visitedPages = storedPages;
  },
  methods: {
    goToPage(page) {
      this.visitedPages.push(page);
      sessionStorage.setItem('visitedPages',
JSON.stringify(this.visitedPages));
      this.$router.push(`/${page.toLowerCase()}`);
    }
  }
};
</script>
```

(a) The list of visited pages will reset to empty after the page is refreshed.

(b) The list of visited pages will be updated with the current page after everynavigation and will not reset even after reload.

(c) The navigation history will not be stored in sessionStorage.

(d) All of these

[Source ans: b]

## 2025 Aug03 Quiz-2

**7.** [Source Q4; MSQ]

Which of the following statement(s) about Web Storage APIs is/are correct?

(a) The `localStorage` object stores data with no expiration date, even after the browser is closed.

(b) The `sessionStorage` object keeps data only for the duration of the page session.

(c) Both `localStorage` and `sessionStorage` can store data in key-value pair format.

(d) Data stored in `sessionStorage` is available across different browser tabs.

[Source ans: a, b, c]

**8.** [Source Q5; MSQ]

Which of the following statement(s) is/are true regarding Single Page Applications (SPA) and Progressive Web Apps (PWA)?

(a) SPAs use client-side routing to dynamically update content without full pagereloads.

(b) PWAs can work offline using service workers.

(c) SPAs always require a server-side language to handle routing.

(d) PWAs can be added to a device's home screen like native apps.

[Source ans: a, b, d]

## 2025 Aug31 ET

**9.** [Source Q16; MCQ]

Consider the following JavaScript code running in a browser:

```
localStorage.setItem('score', '10');
sessionStorage.setItem('multiplier', '3');

for (let i = 0; i < 2; i++) {
    let score = localStorage.getItem('score');
    let multiplier = sessionStorage.getItem('multiplier');

    score = score + 5;
    multiplier = multiplier * 2;

    localStorage.setItem('score', score);
    sessionStorage.setItem('multiplier', multiplier);
}

console.log(localStorage.getItem('score'));
console.log(sessionStorage.getItem('multiplier'));
```

What will be the output in the browser console?

(a)

```
20
12
```

(b)

```
20
NaN
```

(c)

```
NaN
12
```

(d) None of these

[Source ans: d]

## 2025 Oct26 Quiz-1

**10.** [Source Q10; MSQ]

Consider the following HTML and JavaScript code.

HTML file

```
<button id="btn">Click Me</button>

<script>
   const btn = document.getElementById('btn');
   btn.addEventListener('click', () => console.log('A'));
   btn.onclick = () => console.log('B');
   btn.addEventListener('click', () => console.log('C'));
   btn.click();
</script>
```

Among the following console log statements, which ones will be logged when the code executes?

*(Select **all** that apply, irrespective of the order.)*

(a) A

(b) B

(c) C

(d) None

[Source ans: a, b, c]

## 2025 Dec18 ET

**11.** [Source Q17; MCQ]

Consider the following JavaScript code snippet where a web application manages user preferences using localStorage. A user opens the application, sets their theme preference to ”dark”, refreshes the page once, closes the browser tab, and opens a new tab to reload the page. What will be the behavior of the theme preference?

Code:

```
const setTheme = (theme) => {
   localStorage.setItem('theme', theme);
   console.log('Theme set to:', theme);
};

const getTheme = () => {
   return localStorage.getItem('theme') || 'light';
};

setTheme('dark');
console.log(getTheme());
```

(a) The theme will be lost because closing the browser tab clears localStorage

(b) The theme will be preserved as ”dark” because localStorage persists across browser sessions

(c) The theme will be undefined because localStorage is cleared on refresh

(d) The theme will be reset to ”light” when a new tab is opened

[Source ans: b]

**12.** [Source Q23; MCQ]

A developer uses sessionStorage to store authentication tokens in a web application. The user logs in, the token is stored, and then they close the browser window completely. When they reopen the browser later, what will happen? Code:

```
const saveToken = (token) => {
   sessionStorage.setItem('authToken', token);
};

const getToken = () => {
   return sessionStorage.getItem('authToken') || null;
};
```

(a) The token will be preserved and available in the new browser session.

(b) The token will be cleared because sessionStorage is cleared when the browser is closed.

(c) The token will be partially available but require re-authentication.

(d) The token will move to localStorage automatically.

[Source ans: b]

## 2026 Mar15 Quiz-1

**13.** [Source Q4; MCQ]

Consider the following JavaScript code:

```
function addItemToCart(item) {
      let cart = JSON.parse(localStorage.getItem('cart')) || [];
      cart.push(item);
      localStorage.setItem('cart', JSON.stringify(cart));
    }

function getCartItems() {
      return localStorage.getItem('cart') || [];
    }

    addItemToCart({ id: 1, name: 'Laptop' }.name);

    console.log(getCartItems());
```

The above code is initially loaded in the browser, and then the browser is refreshed two times. What will be the final output printed in the console?

(a) `[]`

(b) `['Laptop', 'Laptop', 'Laptop']`

(c) `[{ id: 1, name: 'Laptop' }]`

(d) `[undefined]`

[Source ans: b]

**14.** [Source Q5; MCQ]

Match the following storage mechanisms used in the browser with their correct characteristics.

| Storage Type | Characteristic |
|---|---|
| 1. Session Storage | A. Data automatically encrypts itself without developer intervention. |
| 2. Token Storage | B. Data persists only until the browser/tab is closed. |
| 3. Cookie Storage | C. Often used to store authentication tokens securely on the client side. |
| | D. Data persists across sessions, typically stored on the client and sent with every HTTP request. |

Which of the following is the correct matching?

(a) 1 → B, 2 → C, 3 → A

(b) 1 → B, 2 → A, 3 → C

(c) 1 → A, 2 → C, 3 → B

(d) 1 → B, 2 → C, 3 → D

[Source ans: d]

## 2026 Apr06 Quiz-2

**15.** [Source Q8; MCQ]

Consider the following HTML:

```
<button id="btn1" data-action="delete" data-id="42">Delete</button>
```

And JavaScript code:

```
const btn = document.getElementById('btn1');
const id = btn.dataset.id;
const action = btn.getAttribute('data-action');
```

What are the values of id and action respectively?

(a) id = 42 (number), action = ”delete” (string)

(b) id = 42 (string), action = ”delete” (string)

(c) id = undefined, action = undefined

(d) id = 42 (number), action = undefined

[Source ans: b]

**16.** [Source Q9; MCQ]

Which of the following statements best describes a permalink?

(a) A temporary URL generated for a single browsing session

(b) A stable URL that uniquely identifies a specific resource and is not expected to change

(c) A human-readable URL used only for navigation menus

(d) A URL that depends on the user’s login session or device

[Source ans: b]

## 2026 May06 ET

**17.** [Source Q16; MSQ]

In a Single Page Application (SPA), what are the benefits of client-side routing over server-side routing?

(a) Slower navigation between pages

(b) Reduced load on the backend server

(c) Better SEO performance by default

(d) Seamless page transitions without full reloads

[Source ans: b, c]

## 2026 Jul16 Quiz-1

**18.** [Source Q1; MSQ]

Consider running the following in Browser console:

```
console.log(typeof setTimeout)
console.log(typeof document)
```

Which of the following statements are correct?

(a) The first line prints `"function"` .

(b) The second line prints `"undefined"` .

(c) `document.querySelector()` will work in Browser by default because V8 provides the DOM.

(d) Different runtimes (browser, Node.js, etc.) can provide different APIs even if they use the same JavaScript engine.

[Source ans: a, d]

**19.** [Source Q2; MSQ]

Which of the following are advantages of component-based frontend frameworks like Vue.js?

(a) Code reusability

(b) Better maintainability

(c) Automatic SQL query optimization

(d) Easier UI management

[Source ans: a, b, d]

## 2026 Aug14 Quiz-2

**20.** [Source Q1; MCQ]

A developer opens the Node.js REPL and runs the following line:

```
const heading = document.querySelector("h1")
```

Why does this fail?

(a) Node.js cannot execute JavaScript from the terminal.

(b) The Node.js runtime does not provide browser DOM APIs like `document`

(c) `document.querySelector()` works only after using `setTimeout()`

(d) V8 can execute only arithmetic expressions, not DOM code.

[Source ans: b]

**21.** [Source Q2; MCQ]

Consider the following javascript code:

```
function addProduct(product) {
    let products = JSON.parse(localStorage.getItem('products')) || [];
    products.push(product);
    localStorage.setItem('products', JSON.stringify(products));
}

function getProducts() {
    return JSON.parse(localStorage.getItem('products')) || [];
}

addProduct({ id: 3, name: 'Tablet' }.name);
console.log(getProducts());
```

The above code is initially loaded on the browser and then the browser is refreshed two times. What will be the final output?

(a) ['Tablet', 'Tablet']

(b) ['Tablet', 'Tablet', 'Tablet']

(c) [{ id: 3, name: 'Tablet' }]

(d) [undefined]

[Source ans: b]

**22.** [Source Q8; MSQ]

Suppose the following HTML exists:

```
<div class="card" data-price="499">
  <span id="title">Vue Book</span>
</div>
```

Which of the following statements are correct?

(a) `document.querySelector(".card").dataset.price` returns `"499"` .

(b) `Number(document.querySelector(".card").dataset.price)` converts the value to numeric `499` .

(c) `document.getElementById("title").innerText` can read the visible text `"Vue Book"` .

(d) `document.querySelectorAll(".card")` returns a single element, not a collection.

[Source ans: a, b, c]

## 2026 Sep13 ET

**23.** [Source Q16; MSQ]

Which of the following is/are the advantage(s) of client-side storage?

(a) Saving web generated documents for offline use.

(b) The data can be stored across devices.

(c) Storing personalized site preferences.

(d) IndexedDB API is a kind of client-side storage which can be used for audio/video files.

[Source ans: a, c, d]